

Squaring the Circle by Measurement: The BBP Formula as a Read-Aperture, and Integer-Relation Finding as Physical Relaxation to a Ground State

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June 2026

Abstract

Squaring the circle — constructing, with compass and straightedge, a square equal in area to a given circle — was proven impossible in 1882, when Lindemann showed that π is transcendental. This paper argues that the impossibility belongs to the *operator*, not to π . Construction is the wrong tool; measurement is the right one, and a single formula already demonstrates this. The Bailey–Borwein–Plouffe (BBP) formula, discovered in 1995, extracts the n -th hexadecimal digit of π directly, without computing any preceding digit. We show that, read structurally, BBP is not a series that builds π but a square measuring frame laid over a base-16 grid: its four terms form a closed loop whose straight (radial) part cancels exactly to zero and whose curved (angular) part survives as exactly π . We verify this decomposition symbolically and to fifty digits of precision. We then address why the formula has exactly four terms, replacing earlier claims of “causal necessity” with a falsifiable account: four is what survives elimination, because a three-term frame leaks and a five-term frame is redundant. Finally, we identify the physical principle underneath the algorithm (PSLQ) that discovered the formula. PSLQ does not search; it relaxes a lattice to its lowest-energy configuration, in the same family as the Euclidean algorithm, Gauss reduction, and LLL. We prove that the BBP relation is a genuine energy minimum — a basin — by perturbing it in sixteen directions and finding every one uphill, by relaxing the lattice “cold” (with no answer supplied) and recovering the formula exactly, and by showing the relation is primitive. The same procedure relaxes π^2 into its own sparse survivor, so the mechanism is general. The conclusion is a method with a built-in test — query an invariant against a grid, relax, read the survivor, and verify it as a basin — and a claim it supports: integer-relation discovery obeys the same relaxation logic as crystals,

proteins, and settling fluids. All computational claims were executed; results and code are summarized in an appendix.

1. Construction versus Measurement

For two millennia, computing π meant constructing it. Archimedes inscribed and circumscribed polygons, adding sides to squeeze the circle between bounds. By the fifth century, Zu Chongzhi reached seven decimal places with a 12,288-sided polygon. The seventeenth and eighteenth centuries replaced geometry with analysis — the Gregory–Leibniz series, Machin-like arctangent formulas — and the twentieth century added the rapidly converging series of Ramanujan and the Chudnovskys, the engines behind today's trillion-digit records. Different as these are, they share one architecture: they are state-dependent. To know the n -th digit, you must compute, or pass through, the digits before it. Each is a forward march that carries the entire history of the calculation forward on its back.

Lindemann's 1882 proof that π is transcendental — that it satisfies no polynomial equation with rational coefficients — closed the classical question of squaring the circle. Compass-and-straightedge construction can reach only algebraic numbers (the solutions of such polynomials), so a transcendental length cannot be built by any finite sequence of straight-line steps. The construction is impossible. This paper takes that result as a starting point rather than an endpoint, and asks a narrower question: if π cannot be *constructed*, can it be *measured*? The distinction matters, because measurement and construction are different operations. Construction asks "what is the value, built up from nothing?" Measurement asks "what is at this location?" The first must walk; the second only has to read.

2. BBP Reads π at a Location

In 1995, Simon Plouffe — using the PSLQ integer-relation algorithm discussed in Section 6 — found a formula that broke the forward-march architecture entirely:

$$\pi = \sum_{k=0}^{\infty} 16^{-(k)} \cdot [4/(8k+1) - 2/(8k+4) - 1/(8k+5) - 1/(8k+6)]$$

This series computes the n -th hexadecimal digit of π directly, with almost no memory, without computing any digit before it. The standard description calls it a "spigot" algorithm. That is accurate, and it understates what is happening. A method that returns the n -th digit without the first $(n-1)$ is not generating π — it is *addressing* it. The digits sit at fixed coordinates, and the formula is a read-head that drops onto a chosen coordinate and returns the value there. What goes in is a location, the index n . What comes out is one of sixteen hexadecimal values.

This is not a theoretical reframing; it has an operational fingerprint at the largest scales. When Shigeru Kondo and Alexander Yee set a world record by computing π to five trillion decimal digits with the sequential Chudnovsky algorithm, the forward build took roughly ninety days. To verify the result, they

ran BBP to extract the base-16 digits at the far end of the five-trillion-digit sequence directly. That confirmation took about 64 hours rather than another ninety days, because the read-head can be dropped trillions of places into π independently of every digit around it. The digits behave as addressable coordinates, not as the cumulative output of a walk. That is the empirical sense in which BBP measures rather than constructs.

3. The Mechanism: A Square Measuring a Circle

To see *how* the read-head works — not what it does, but the mechanism that produces the digit — the discrete series is best rewritten as its continuous analog. Beginning over the interval $[0, 1/\sqrt{2}]$ and substituting $y = \sqrt{2} \cdot x$ normalizes the bounds to the unit interval and collapses the whole operation into a single rational integral:

$$\pi = \int_0^1 16(4 - 2y^3 - y^4 - y^5) / (16 - y^8) dy$$

This is the geometric mold of the operation, and it has two parts with two jobs. The denominator, $(16 - y^8)$, is the base-16 grid — an eight-fold structure (a 2^4 lattice) whose eight roots set the geometry. The numerator, $16(4 - 2y^3 - y^4 - y^5)$, is the footprint of the four-term mask: the polynomial stamped onto that grid by the four active terms $\{1, 4, 5, 6\}$ with their weights $[4, -2, -1, -1]$.

3.1 The Zero-Sum Mask Is a Closed Loop

The mask weights sum to exactly zero: $4 - 2 - 1 - 1 = 0$. In experimental mathematics, a BBP-type series whose coefficients sum to zero is classified as a “null” series and treated as a bookkeeping curiosity. Geometrically it means something specific. When a surveyor measures a parcel by walking its perimeter, the net displacement must return to zero, or the boundary did not close. A square is a four-sided figure that closes on itself. The mask is the straight scaffold of a measuring frame, and because its weights sum to zero, the steps it takes along the radial boundary return to the origin. The frame closes — it measures itself out — and the four active terms are its four corners.

3.2 Factoring: What the Frame Sheds

Factoring the integrand confirms the closure exactly. Over the real numbers the denominator splits into four irreducible quadratics:

$$16 - y^8 = -(y^2-2)(y^2+2)(y^2-2y+2)(y^2+2y+2)$$

These four quadratics are the entire geometry of the grid — but not all of it bears on π ; some roots are silent, setting the lattice without contributing to the final residue. The numerator, factored, is:

$$16(4 - 2y^3 - y^4 - y^5) = -16(y-1)(y^2+2)(y^2+2y+2)$$

Numerator and denominator share the factor $(y^2+2)(y^2+2y+2)$. On simplification it cancels completely — the mask is precisely shaped to shed the silent, non-contributing roots — leaving the minimal, irreducible form (verified by symbolic computation):

$$4y/(y^2-2) - 4(y-2)/(y^2-2y+2)$$

This split is the physical bifurcation of the read-head into two channels: a radial channel (the straight scaffold) and an angular channel (the curve).

3.3 Radial Cancels, Angular Survives

Integrating the two channels across $[0,1]$ shows the mechanism in one image. The radial channel, $4y/(y^2-2)$, integrates to a logarithm — logarithms being the mathematics of straight, radial scaling. Its pieces evaluate to $-2 \cdot \log(2)$ and, from the second channel's logarithmic part, $+2 \cdot \log(2)$. They are equal and opposite:

$$\Delta L = -2 \cdot \log(2) + 2 \cdot \log(2) = 0$$

The straight scaffold annihilates itself across the boundary and leaves no residue, exactly as the zero-sum mask predicted. What remains is the part straight edges cannot account for — the curvature — carried by the arctangent, the operator of phase and rotation:

$$\Delta \Theta = 4 \cdot \arctan(0) - 4 \cdot \arctan(-1) = 0 - (-\pi) = \pi$$

That is the whole machine. The square closes and cancels ($\Delta L = 0$); the circle it could not capture is left standing in the gap ($\Delta \Theta = \pi$). BBP does not build π . It builds a zero-sum box and reads the curvature trapped inside it. This is squaring the circle — not by construction, which Lindemann forbids, but by measurement, which the closed frame makes trivial.

Verified: the symbolic factorizations and partial-fraction split above were confirmed by computer algebra; $\Delta L = 0$ and $\Delta \Theta = \pi$, and the four-term series equals π , were confirmed to 50 digits of precision.

4. Why Exactly Four Terms

It is tempting to call the four-term structure a necessity — to say the universe demands four corners. That language overstates what can be shown, and the honest account is both weaker and more useful: four is not forced; it is what survives when the alternatives are eliminated. The question is not “why must it be four?” but “what is left standing after three- and five-term frames fail?”

4.1 Three Terms Leak

To isolate the arctangent that yields π , the numerator must reproduce and cancel the silent factor $(y^2+2)(y^2+2y+2)$. A three-term mask — three active terms, three weights — lacks the degrees of freedom to generate that factor. The silent roots fail to cancel; the simplified form retains complex quadratic

denominators; integrating them produces logarithmic terms across multiple irrational phases that no longer balance to zero. The radial part leaks, the loop does not close, and π cannot be cleanly isolated. A three-term frame in this grid leaks, and is eliminated.

4.2 Five Terms Are Redundant

A five-term mask fails the opposite way. The grid admits known “zero relations” — valid series whose weights sum to zero and whose total value is also exactly zero. These are closed loops that enclose no area; they measure nothing. Given five degrees of freedom to express π , the algorithm does not discover a richer shape — it takes the working four-term structure and superimposes a zero relation on top of it. The fifth term contributes nothing to the measurement; it adds redundant, self-cancelling steps. (In practice, researchers searching for genuinely new formulas must deliberately exclude these zero relations from the search to prevent exactly this padding.) A five-term frame is redundant, and is eliminated.

4.3 Four Survives

Three leaks; five pads. Four has precisely enough freedom to shed the silent roots without triggering the redundant zero relations — enough to cancel the radial flow in both directions and leave the clean arctangent that yields π . Four is what remains when leakage and redundancy are removed. The frame is square because four is the minimal closed orthogonal corner-count that fits this grid. It is not magic and it is not decreed. It is the survivor.

This establishes four as a minimal survivor — three-term frames leak, five-term frames pad. It does not claim four is the unique survivor across every unexplored region of the search space; that stronger statement is left open.

5. The Grid Sets the Corner Count

That the term count is set by the grid, not chosen, is confirmed by the same kind of formula in other bases, each discovered independently. Each base demands a different scaffold to force radial cancellation, and the corner counts fall out of the geometry rather than being selected. The pattern is not even monotonic — which is what an honest structural law looks like, reporting the data rather than forcing a clean line through it.

Formula	Base / Grid	Active Terms	Corners	Reading
Adamchik–Wagon	4 (2^2)	{1,2,3}	3	Triangle on a compressed grid; the smaller space drops the freedom to three.
Plouffe (BBP)	16 (2^4)	{1,4,5,6}	4	Square; the orthogonal frame that fits the 2^4 grid with minimal redundancy.

Bellard	1024 (2 ¹⁰)	{1,3,5,7,9}+	7	Seven-cornered frame; faster per term, but heavy scaffold to cancel a large leap.
Huvent	64 (2 ⁶)	8 residues	8	Eight-cornered frame to force closure in base-64.

Base-4 shows that fewer terms are possible when the grid permits — a three-cornered scaffold closes the loop in the compressed space. Bellard’s base-1024 shows the converse: advancing ten binary digits per term generates enormous radial leakage, and the scaffold must grow to seven corners with heavily scaled fractions to cancel it. Among them, base-16 is structurally clean — the base is 2⁴ and the mask needs exactly four terms — so the scale law and the aperture map orthogonally onto each other. As a general principle, the frame must be exactly as complex as the interference it has to cancel: a purely radial constant such as log(2) needs only a single term, while extracting $\sqrt{5} \cdot \log(\phi)$ requires a frame of roughly forty terms to cancel the deep Fibonacci–Lucas structure embedded in the golden ratio. Term count is set by the depth of interference, not by choice.

6. What the Algorithm Is Doing: Relaxation, Not Search

The formula was found by PSLQ, an integer-relation algorithm that, given a list of real numbers, finds the integer combination of them that equals zero, if one exists. It is worth asking what PSLQ physically *does*, because the answer explains why the four-term frame is special and connects the whole construction to physics. PSLQ does not enumerate candidate relations. It builds a lattice from the input numbers and reduces it — repeatedly subtracting whole-number multiples to shrink the structure, rotating to expose the next reduction — until a short relation emerges as the survivor. That reduction primitive is old. PSLQ (1992) refines LLL (1982); LLL generalizes Gauss’s two-dimensional lattice reduction (c. 1800); and Gauss’s method is the Euclidean algorithm (c. 300 BC) lifted from integers to vectors. Every link performs the same act: subtract the largest admissible whole multiple, keep the residue, repeat until the state stops moving.

That act is not calculation. It is relaxation — a system settling to its lowest-energy state. Run a Euclidean or Gauss reduction and track the size of the state as an energy: it decreases monotonically and stops at a floor, exactly as a crystal settles into its lattice, a protein folds into its basin, or water finds its level. PSLQ is this same settling, carried to its mature form. The relation it returns is the configuration the lattice falls into when it can fall no further.

7. The BBP Relation Is a Ground State

If PSLQ is genuinely relaxation, then the BBP relation should be a genuine energy minimum — a basin you fall into and cannot climb out of cheaply — not merely a relation that happened to be returned. This is directly testable. Define the grid sums and an energy for any candidate mask:

$$S_{-j} = \sum_k 1/(16^k (8k+j)) \quad \text{energy}(\text{mask}) = |\pi - \sum_j \text{mask}_{-j} \cdot S_{-j}|$$

The ground state is energy zero. The BBP mask places weights [4, -2, -1, -1] on terms {1, 4, 5, 6} and has energy 2×10^{-50} — zero to working precision, sitting on the floor. Perturbing it tests whether the floor is a basin: change each of the eight weights by ± 1 and recompute the energy. All sixteen directions rise.

Perturbation from BBP	Resulting energy	Direction
term 1 weight ± 1 (the +4 drive)	1.0072	uphill — steepest
term 2 weight ± 1	0.5065	uphill
term 3 weight ± 1	0.3392	uphill
term 4 weight ± 1	0.2554	uphill
term 5 weight ± 1	0.2050	uphill
term 6 weight ± 1	0.1713	uphill
term 7 weight ± 1	0.1472	uphill
term 8 weight ± 1 (empty tail)	0.1291	uphill — softest

Every neighbor is higher; there is no free sideways move. By the operational definition, the BBP relation is a basin — a ground state. Two further checks confirm it is the right minimum. Relaxing the lattice cold — supplying PSLQ only π and the eight grid sums, with no information about the answer — returns the BBP mask exactly, [1, -4, 0, 0, 2, 1, 1, 0, 0] on $(\pi, S_1, \dots, S_8)$. The relation is where the lattice settles; nothing placed it there. And it is primitive: doubling the mask leaves the floor (it evaluates to π , not 0), so only the primitive relation reaches zero.

Verified to 50 digits: BBP energy $\approx 2 \times 10^{-50}$; all sixteen single-weight perturbations increase the energy (a strict basin); cold relaxation returns the BBP mask exactly; doubling the mask leaves the floor (primitive).

7.1 The Shape of the Basin

The basin is not a symmetric bowl, and the asymmetry is the point. Its walls differ in steepness: the term carrying the +4 drive is the steepest wall (an uphill step near 1.0), while the empty high-index tail is the softest (near 0.13) — the deepest part of the well anchors to the corner carrying the most weight. The closure has a definite shape as well. The weights are +4 on one term and -2, -1, -1 on the other three, and the way they sum to zero is specific: $4 = 2 + 1 + 1$. One positive corner exactly balances three negative ones. The frame does not close by four equal sides; it closes by a one-against-three balance — a drive against its distributed counter-flow. Placed by angle on the eight-fold grid, the four terms sit at 45° , 180° , 225° , and 270° : the drive and its heaviest counter-term are opposite, but the remaining two are bunched in one quadrant. The configuration leans; it is not a symmetric cross. A perfectly symmetric mask would cancel to nothing, so the basin exists precisely because it leans.

8. A General Method, and Its Built-In Test

A single basin around π would show only that π is special. The mechanism is general, and two checks confirm it. Relaxing π^2 against a squared grid — terms $1/(16^k (8k+j)^2)$ — settles cold into its own sparse survivor, with residual on the order of 10^{-49} : a clean ground state for π^2 on its own grid. And the grid is not uniquely privileged at one offset; shifting it from $8k+1\dots+8$ to $8k+2\dots+9$ still yields a ground-state relation when relaxed. π and π^2 both fall out of their grids as ground-state survivors by the same cold relaxation, which makes this a method rather than a coincidence of one constant.

The method is: supply an invariant and a grid of candidate terms, relax the associated lattice, and read the sparse survivor — the relation falls out as the ground state. It is not lookup, because the answer is never supplied, and it is not coincidence, because the survivor is a basin. The method comes with its own verification. To confirm that a returned relation is genuine and not an artifact, perturb it and check that every direction raises the energy; a true survivor is a basin, a false one is not. This turns the loose phrase “the answer is what survives elimination” into a runnable test: elimination is the climb out of the well, and the answer is the floor.

A worked negative, recorded for honesty: a separate attempt to read “invariant angle ratios” by wrapping π , φ , and e on a spiral produced ratios near $8/7$ and $2/3$ that did not survive testing — they were artifacts of an unwrapped regime, were loose fits (about 1% and 16% off), and the integer-relation test returned no relation. The branch is recorded as a dead end. It narrowed the search; that is its value.

9. Measurement and Creation as One Event

With the relation proven to be a ground state, the two readings of BBP cease to be a choice of interpretation and become two ends of one physical event. Read one direction, the relaxation is measurement: a numerical field exists, and the settle reads its invariant. Read the other direction, it is creation: a grid is supplied as a constraint, and the admissible object is whatever survives the settle — π rendered as the ground state of that grid. These are not two mechanisms but one downhill roll seen from opposite sides. BBP read one way measures π ; read the other way it supplies the access geometry in which π becomes locally addressable. The lattice does not care which end one stands at; it settles, and the survivor is at once the measurement and the frame.

This is the precise sense in which mathematical relation-finding is bound by the same physics as matter. The relation is not found by a search that floats above the world. It is found by the same relaxation that folds a protein, settles a crystal, or finds the level of water. The relation is the stable configuration; the algorithm is the relaxation engine; least action is the floor beneath both. The impossibility of squaring the circle was never a property of π . It was a property of construction — the wrong operator. Under measurement, realized as a closed frame relaxed to its ground state, the circle is squared, the straight scaffold cancels, and π is read, digit by digit, from the gap.

10. Conclusion

Squaring the circle by forward construction is impossible, not by any property of the universe but because construction is the wrong operator for a transcendental target. Measurement is the right operator, and BBP is the worked instrument. Structurally it is a square frame on a base-16 grid: a four-term closed loop whose weights sum to zero, whose radial part cancels to exactly zero, and whose angular part survives as exactly π — all verified symbolically and to fifty digits. Four terms is not forced; it is what survives when three-term frames leak and five-term frames pad, and the corner count is set by the grid, as independent formulas in other bases confirm. The algorithm that found the formula does not search; it relaxes a lattice to its ground state, in the lineage of Euclid, Gauss, and LLL. The BBP relation is a proven basin — every perturbation climbs, it falls out of a cold relaxation, and it is primitive — and its basin is an asymmetric, leaned frame rather than a symmetric cross, which is why it exists rather than cancelling. The mechanism generalizes to π^2 and to shifted grids, giving a method with a built-in test: query an invariant against a grid, relax, read the survivor, and verify it as a basin. The broader statement this supports is that integer-relation discovery obeys the same relaxation logic as crystals, proteins, and settling fluids — that in this precise operational sense, mathematics is not outside physics. The answer is the floor.

Appendix. Computational Summary

All quantitative claims in this paper were executed; the following summarizes what was run and what was returned.

(A) Channel split (Section 3). The four-term BBP series, summed to 200 terms at 50-digit precision, equals π (difference $\approx 3 \times 10^{-50}$). Symbolic computer algebra confirms $16-y^8 = -(y^2-2)(y^2+2)(y^2-2y+2)(y^2+2y+2)$, confirms the numerator factors as $-16(y-1)(y^2+2)(y^2+2y+2)$, confirms the shared GCD $(y^2+2)(y^2+2y+2)$ cancels, and confirms the partial-fraction split $4y/(y^2-2) - 4(y-2)/(y^2-2y+2)$. Integrated over $[0,1]$: $\Delta L = -2\log 2 + 2\log 2 = 0$ and $\Delta\Theta = -4 \cdot \arctan(-1) = \pi$.

(B) Ground-state / basin (Section 7). With $S_j = \sum 1/(16^k(8k+j))$ and $\text{energy}(\text{mask}) = |\pi - \sum \text{mask}_j S_j|$: the BBP mask has energy $\approx 2 \times 10^{-50}$; all sixteen single-weight ± 1 perturbations increase the energy (values from ≈ 0.13 to ≈ 1.01); a cold PSLQ relaxation of $(\pi, S_1 \dots S_8)$ returns $[1, -4, 0, 0, 2, 1, 1, 0, 0]$ exactly; doubling the mask gives energy $\approx \pi$ (so the relation is primitive).

(C) Generalization (Section 8). A cold relaxation of $(\pi^2, T_1 \dots T_8)$ with $T_j = \sum 1/(16^k(8k+j)^2)$ returns a sparse relation with residual $\approx 10^{-49}$. A relaxation of $(\pi, S_2 \dots S_9)$ on the offset grid returns a ground-state relation. Tools: Python with mpmath (50-digit precision) and SymPy (symbolic factorization).